

Christian H. Johansen

DATA SCIENCE · BIOINFORMATICS

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PhD student applying deep learning methods to integrate omics and health registry data for predictive modeling. Dedicated to creating robust, reproducible software solutions that can deliver translational impact and real-world healthcare advancements.

Experience

Novo Nordisk Foundation Center for Protein Research, University of Copenhagen

Copenhagen, Denmark

PHD STUDENT

Aug. 2023 - PRESENT

- Research on the integration and representation of large, longitudinal electronic health record datasets for deep learning applications.
- Developed a **transformer-based neural network detection tool** for young-onset type 2 diabetes, integrating primary and secondary care data.
- Created a novel embedding approach of laboratory tests and trained **clinical foundation models**.

University of Exeter

Exeter, United Kingdom

VISTING RESEARCHER

Mar. 2025 - Aug. 2025

- Generated genetic risk scores for type 1 and type 2 diabetes and integrated them into a deep learning model to stratify adult-onset diabetes cases at the time of diagnosis.

Novo Nordisk Foundation Center for Protein Research, University of Copenhagen

Copenhagen, Denmark

RESEARCH ASSISTANT

Aug. 2022 - Aug. 2023

- Led multi-omics integration efforts on a cross-European research project integrating high-dimensional multi-omics data from newly diagnosed type 1 diabetes patients to model future C-peptide trajectories, supporting targeted clinical trial designs.

Skills

Programming

Python, R, SQL, Bash, Rust, LaTeX

Data Science & Analytics

Pytorch, Scikit-learn, Numpy, Polars, Pandas, PostgreSQL, DuckDB, Hugging Face

Visualizations

Shiny, Tidyverse, Dash, Tensorboard

Devops & Computing

Git, SLURM, Docker, Snakemake

Languages

Danish, English, German

Education

Technical University of Denmark

Kongens Lyngby, Denmark

M.SC.ENG IN BIOINFORMATICS AND SYTEMS BIOLOGY (GPA: **11.3/12**)

Sep. 2020 - Jul. 2022

- Statistical modelling, machine learning, data science, algorithms in bioinformatics, Next-generation sequencing and Immunology.
- Thesis on improving T cell specificity predictions using deep learning at Morten Nielsen's Lab.

University of Copenhagen

Copenhagen, Denmark

B.SC. IN BIOCHEMISTRY (GPA: **11.4/12**)

Sep. 2017 - Jun. 2020

- Protein interactions, immunology and structural biology.
- Developing a prokaryotic 2-hybrid system at Martin Willemoes Lab

Publications

2025 (in press)	Detection of Young-Onset Type 2 Diabetes Using Deep Learning Across Primary and Secondary Care: a nationwide retrospective cohort study	<i>Lancet Digital Health</i>
2025	Subgrouping patients with ischemic heart disease by means of the Markov cluster algorithm	<i>Communications Medicine</i>
2024	Multi-omics analysis reveals drivers of loss of β-cell function after newly diagnosed autoimmune type 1 diabetes: An INNODIA multicenter study	<i>Diabetes / Metabolism Research and Review</i>

Presentations

9th Annual Danish Bioinformatics Conference	<i>Aalborg, Denmark</i>
A DEEP LEARNING APPROACH FOR EARLY DETECTION OF YOUNG-ONSET TYPE 2 DIABETES	<i>Oct. 2025</i>
Intelligent Systems for Molecular Biology (ISMB) 2024	<i>Montreal, Canada</i>
PREDICTIVE SCREENING OF TREATMENT NEEDS FOR YOUNG-ONSET TYPE 2 DIABETES	<i>Jul. 2024</i>

Teaching

Bioinformatics and Systems Biology	<i>Copenhagen, Denmark</i>
HUMAN BIOLOGY	<i>Sep. 2022 - Sep. 2025</i>
• Lead hands-on exercises designed to build fundamental understanding of machine learning concepts through experimentation and visual evaluation of models.	
Big Data Analytics and ML	<i>Copenhagen, Denmark</i>
BRIDGE - TRANSLATIONAL EXCELLENCE PROGRAMME	<i>Oct. 2025</i>
• Delivered lectures on clinical deep learning use cases, emphasizing best practices in machine learning, including data leakage prevention, validation strategies, and model interpretability. • Designed exercises for implementing clinical prediction models and applying core machine learning concepts such as feature engineering, model evaluation, and performance interpretation.	

Certificates

ISO 21500, ISO 21502	<i>Dansk Standard</i>
PROJECT MANAGEMENT	<i>Jul. 2021</i>
• Knowledge of high-level project management principles and terminology • Understanding of practical project delivery guidance, including governance, tailoring, and benefits realization within organizational contexts.	